# BYOUNGUK MIN

Department of Computer Science, Purdue University, West Lafayette, IN 47907 ⋄ +1 (765) 409-7422
min87@purdue.edu ⋄ byoungukmin.github.io ⋄ Google Scholar ⋄ GitHub

## RESEARCH INTERESTS

Computer architecture and compilers, with a focus on intermittent computing for battery-less energy-harvesting systems: GPUs and nonvolatile processors that stay correct and efficient under frequent power failures.

## EDUCATION

**Purdue University** *Aug. 2021–Present*
Ph.D. in Computer Science (expected Dec. 2027) GPA: 3.58/4.00
Advisor: Prof. Changhee Jung

**Yonsei University** *Mar. 2019–Feb. 2021*
M.S. in Computer Science GPA: 3.89/4.30
Thesis: *PBQueue: Persistent Batched Lock-Free Queue under Persistent Memory*
Advisor: Prof. Bernd Burgstaller

**Sungkyunkwan University** *Mar. 2013–Feb. 2019*
B.S. in Computer Science GPA: 3.67/4.50

## PUBLICATIONS

[1] **Byounguk Min**, Yilun Wu, Yuchen Zhou, Kasım Sinan Yıldırım, Khakim Akhunov, Jianping Zeng, Jongouk Choi, Wenjie Xiong, Dongyoon Lee, and Changhee Jung. "iGPU: Enabling GPUs for Intermittent AI on Batteryless Systems." In *Proceedings of the 59th IEEE/ACM International Symposium on Microarchitecture* (**MICRO'26**), Athens, Greece, Oct. 2026.

[2] Yida Zhang, Shao-Yu Huang, **Byounguk Min**, Andrew Inho Park, Gan Fang, Jongouk Choi, and Changhee Jung. "Achieving Speculative Intermittent Computation on Commodity Energy Harvesting Hardware." In *ACM SIGBED International Conference on Embedded Software* (**EMSOFT'26**), Oct. 2026.

[3] Noureldin Hassan, **Byounguk Min**, Changhee Jung, Yan Solihin, and Jongouk Choi. "WarmCache: Exploiting STT-RAM Cache for Low-Power Intermittent Systems." In *Proceedings of the 52nd ACM/IEEE International Symposium on Computer Architecture* (**ISCA'25**), Tokyo, Japan, Jun. 2025. [DOI]

[4] Yilun Wu, **Byounguk Min**, Mohannad Ismail, Wenjie Xiong, Changhee Jung, and Dongyoon Lee. "IntOS: Persistent Embedded Operating System and Language Support for Multi-threaded Intermittent Computing." In *Proceedings of the 18th USENIX Symposium on Operating Systems Design and Implementation* (**OSDI'24**), Santa Clara, CA, USA, Jul. 2024. [Link]

[5] Shinhyung Yang, Seongho Jeong, **Byounguk Min**, Yeonsoo Kim, Bernd Burgstaller, and Johann Blieberger. "Design-Space Evaluation for Non-Blocking Synchronization in Ada: Lock Elision of Protected Objects, Concurrent Objects, and Low-Level Atomics." *Journal of Systems Architecture* (**JSA**), vol. 110, 101764, 2020. [DOI]

## RESEARCH PROJECTS

**Enabling Heterogeneous Computing Systems** Purdue University
*Supervised by Prof. Changhee Jung* *Dec. 2025–Present*

- Developing a power-aware work-partitioning scheme that splits each inference task across compute engines according to the available energy budget, with idempotent recovery when work is reassigned after a power failure.

**IC-SIM: Intermittent Computing Simulator** Purdue University
*Supervised by Prof. Changhee Jung* *Nov. 2025–Present*

- Building IC-SIM, a simulator for energy-harvesting systems that models power failures, capacitor/battery dynamics, and checkpoint/restart behavior to study end-to-end forward progress under intermittent power.

**iGPU: Enabling GPUs for Intermittent AI on Batteryless Systems** Purdue University
*Supervised by Prof. Changhee Jung* *Nov. 2022–Aug. 2026*

- Designed a GPU architecture for intermittent AI inference on batteryless energy-harvesting systems, a domain where prior architecture work has focused on CPUs.
- First-author paper published at MICRO'26.

**WarmCache: STT-RAM Cache for Intermittent Systems** Purdue University
*Supervised by Prof. Changhee Jung* *Nov. 2024–Jun. 2025*

- Contributed to WarmCache, an STT-RAM cache design with relaxed non-volatility for power-hungry energy-harvesting systems; published at ISCA'25.

**IntOS: Embedded OS for Energy-Harvesting Platforms** Purdue University
*Supervised by Prof. Changhee Jung* *Oct. 2023–Mar. 2024*

- Contributed to IntOS, an embedded OS and language support for multi-threaded intermittent computing on battery-less platforms; published at OSDI'24.
- IntOS offers conventional threads and transactions, with automatic undo-logging of persistent objects in non-volatile memory.

**PBQ: Persistent Lock-Free Queue for NVM** Yonsei University
*Supervised by Prof. Bernd Burgstaller* *Jan. 2020–Jun. 2020*

- Designed PBQ, a persistent lock-free queue that uses batching to achieve both high throughput and durability on non-volatile memory.
- Evaluated on an Intel Optane DC persistent memory server and a 120-core manycore server.

**Relaxed Memory Consistency Evaluation on RISC-V** Yonsei University
*Supervised by Prof. Bernd Burgstaller* *Jan. 2020–Apr. 2020*

- Extended the Ada concurrent-object work to RISC-V, evaluating non-blocking concurrent objects built on its atomic read-modify-write operations.
- Measured scalability of non-blocking algorithms on a 4-core SiFive HiFive Unleashed board.

**Non-Blocking Synchronization in the Ada Programming Language** Yonsei University
*Supervised by Prof. Bernd Burgstaller* *Apr. 2019–Nov. 2019*

- Proposed concurrent objects, a language-level primitive that encapsulates non-blocking synchronization built on atomic read-modify-write operations under relaxed memory models.
- Evaluated on x86 and ARMv8 to characterize how memory consistency models affect performance; published in JSA (2020).

## TEACHING EXPERIENCE

**CS 35200: Compilers: Principles and Practice** — Purdue University
*Head TA (Fall 2026); TA (Fall 2023, 2024, 2025)* — *Instructor: Prof. Changhee Jung*

· Undergraduate compilers course. As Head TA, coordinate the TA team, programming-assignment logistics, and course communications.

**CS 50200: Compiling and Programming Systems** — Purdue University
*TA (Spring 2024, 2025, 2026)* — *Instructor: Prof. Changhee Jung*

· Graduate course on advanced compiler design.

**CSI4104: Compiler Design** — Yonsei University
*TA (Fall 2019)* — *Instructor: Prof. Bernd Burgstaller*

· Programming language design patterns and compiler techniques.

**CSI2100: Computer Programming** — Yonsei University
*TA (Spring 2019, 2020)* — *Instructor: Prof. Bernd Burgstaller*

· Python programming fundamentals and programming techniques.

## TECHNICAL SKILLS

| | |
|---|---|
| **Languages** | C, C++, Python, Ada |
| **Simulators** | gem5, GPGPU-Sim / Accel-Sim |
| **Compilers** | LLVM compiler infrastructure |
| **Systems** | Parallel and concurrent programming; persistent-memory programming (PMDK) |

## MILITARY SERVICE

**Republic of Korea Army, KATUSA** — *Jun. 2015–Mar. 2017*
Korean Augmentation To the United States Army, serving with United States Forces Korea (USFK)